



EC3452 – ELECTROMAGNETIC FIELDS

UNIT 1:

PART-B

- Rectangular, **cylindrical** and spherical **coordinate systems**
- **Divergence theorem**
- **Verify theorems for different path, surface and volume**

PART-A

- **Gradient of a scalar field**
- **Stoke's theorem** / Helmholtz's theorem
- **Curl of a vector field** / Electromagnetic mode

UNIT 2:

PART-B

- **Gauss's law and applications**
- **Poisson's and Laplace's Equations**
- **Capacitance in Parallel / Cylindrical / Spherical Capacitors**
- **Problems: Electrostatics Boundary value**

PART-A

- **Coulomb's law** / Capacitance
- **Kirchhoff's Laws (KVL / KCL)**
- **Electric field** / **Electric Potential** / **Electric flux density**
- **Current density** / **Ohm's law**
- **Electrostatic energy** / **Electromotive force**

UNIT 3:

PART-B

- **Vector Magnetic Potential**





- **Biot-Savart law and applications**
- **Calculation: Magnetic field intensity for various current distributions Magnetic circuits**

PART-A

- **Lorentz force equation / Ampere's law**
- **Inductance and Inductors**
- **Magnetic energy / Magnetic forces and Torques**

UNIT 4:

PART-B

- **Maxwell's Equations**
- **Wave propagation with the aid of Maxwell's equations**
- **Wave Equation For Electric & Magnetic Field**

PART-A

- **Faraday's law / Displacement current**
- **Electromagnetic Boundary Conditions**
- **Time-Harmonic Fields / Maxwell-Ampere law**

UNIT 5:

PART-B

- **Poynting Theorem**
- **Normal incidence at a plane dielectric boundary**

PART-A

- **Group velocity / Poynting Vector**
- **Electromagnetic power flow**
- **Brewster Angle / Skin Depth**
- **Plane waves in lossy media**



IMPORTANT QUESTIONS – UNIT 1

1. Rectangular Coordinate Systems
2. Cylindrical Coordinate Systems
3. Spherical Coordinate Systems
4. Gradient, Divergence, curl and stoke's Theorem
5. Problems



IMPORTANT QUESTIONS – UNIT 2

1. Gauss Law and its Applications
2. Electro static Boundary Conditions
3. Electric field Intensity Expression
4. Poisson's & Laplace Equation
5. Parallel Capacitance , Cylindrical and Spherical Capacitors
6. Problems



IMPORTANT QUESTIONS – UNIT 3

1. Biot Savart law and its applications
2. Ampere's Law
3. Magnetic Field Intensity – Derivation & Calculations
(using Ampere's Law and biot Savart Law)
4. Vector Magnetic Field – Lorentz force Equation
5. Magnetic Forces and Torques
6. Problems



IMPORTANT QUESTIONS – UNIT 4

1. Maxwell's Equations
2. Wave Equations
3. Electromagnetic Boundary Conditions



IMPORTANT QUESTIONS – UNIT 5

1. Plane waves in lossless and lossy dielectric
2. Poynting Vector
3. Normal Incidence (Conducting Boundary & Plane Dielectric Boundary)
4. Problems



IMPORTANT QUESTIONS

PART – A

UNIT – 1



1. State Stoke's theorem
2. State Coulomb's law.
3. State Divergence theorem
4. State Gauss's law in Integral and Differential forms.
5. State Gauss Divergence theorem.

4. State Gauss's law in integral and Differential forms.

5. State Gauss Divergence theorem.

UNIT – 2



1. Write the Poisson's equation and Laplace's equation.

2. Obtain Poisson's equation from Gauss's law.

3. Define dielectric strength.

4. Give the relationship between potential gradient and electric field.

5. What is polarization?

UNIT – 3

1. Write Lorentz force equation.
2. State Biot Savart law
3. State Ampere's circuital law
4. Define Torque and Write its expression.



UNIT – 4

1. Write Maxwells equation.
2. State faraday's law of electromagnetic induction

2. State Biot Savart law
3. State Ampere's circuital law
4. Define Torque and Write its expression.



UNIT – 4

1. Write Maxwells equation.
2. State faraday's law of electromagnetic induction
3. What is displacement current?
4. State Maxwell ampere law.



UNIT – 5

1. Define poynting vector.
2. What is the importance of poynting vector?
3. What is meant by group velocity?

PART – B

UNIT – 4

3. What is meant by group velocity?

PART – B

UNIT – 1



1. State and prove stoke's and divergence theorem.
2. State and prove Helmholtz theorem.

UNIT – 2

1. Explain poisson's and laplace equations.
2. Explain applications of Gauss law.

2. State and prove Helmholtz theorem.

UNIT – 2

1. Explain poisson's and laplace equations.
2. Explain applications of Gauss law.
3. Derive an expression for electric potential.

UNIT – 3

1. State Biot Savart Law. Explain its applications.
2. Derive Lorentz force equation.



UNIT – 4

1. Derive Maxwell's equations.



UNIT – 5

1. Explain poynting vector.
2. Derive an expression for
 - i) Normal incidence at a plane conducting boundary
 - ii) Normal incidence at a plane dielectric boundary.

PART – C

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1. Explain poynting vector.
 2. Derive an expression for
 - i) Normal incidence at a plane conducting boundary
 - ii) Normal incidence at a plane dielectric boundary.



PART – C

1. Derive an expression for capacitance for parallel, cylindrical and spherical capacitors.
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